

# A POLYNOMIAL PARAMETRIZATION OF THE DIOPHANTINE HYPERSURFACE $xyz + t^2 = 1$

ABSTRACT. We give an explicit three-parameter polynomial construction for the equation

$$xyz + t^2 - 1 = 0.$$

For each sign  $\varepsilon \in \{1, -1\}$ , the polynomial map

$$\Phi_\varepsilon(a, b, c) = (a, b, c(2 - abc), \varepsilon(1 - abc))$$

takes integer parameters to integer solutions. Over  $\mathbb{Q}$ , either map is an isomorphism onto the dense open set on which  $xy \neq 0$ , and the two maps together cover every rational solution. We also determine exactly which integral points arise from integral parameters in either chart.

## 1. THE POLYNOMIAL CONSTRUCTION

Let

$$X : \quad xyz + t^2 - 1 = 0$$

be the affine hypersurface in four variables. For a fixed  $\varepsilon \in \{1, -1\}$ , define

$$\Phi_\varepsilon : \mathbb{A}^3 \longrightarrow \mathbb{A}^4, \quad \Phi_\varepsilon(a, b, c) = (a, b, c(2 - abc), \varepsilon(1 - abc)).$$

**Theorem 1.** *For every commutative ring  $R$ , every  $a, b, c \in R$ , and either sign  $\varepsilon \in \{1, -1\}$ , the point  $\Phi_\varepsilon(a, b, c)$  lies on  $X(R)$ . In particular,*

$$\boxed{x = a, \quad y = b, \quad z = c(2 - abc), \quad t = \varepsilon(1 - abc)}$$

*is a three-parameter polynomial family of integer solutions when  $a, b, c \in \mathbb{Z}$ .*

*Proof.* Put  $s = abc$ . For the point given above,

$$xyz = abc(2 - abc) = s(2 - s),$$

whereas

$$t^2 - 1 = (1 - s)^2 - 1 = s^2 - 2s = -s(2 - s).$$

Consequently,

$$xyz + t^2 - 1 = s(2 - s) - s(2 - s) = 0.$$

This is a polynomial identity over  $\mathbb{Z}$ , so it remains valid over every commutative ring. □

The construction is obtained directly from the factorization

$$1 - t^2 = (1 - t)(1 + t).$$

For the positive-sign chart, impose  $1 - t = abc$ . Then  $t = 1 - abc$  and  $1 + t = 2 - abc$ , whence

$$1 - t^2 = abc(2 - abc).$$

Taking  $x = a$ ,  $y = b$ , and  $z = c(2 - abc)$  gives the formula above. Replacing  $t$  by  $-t$  gives the negative-sign chart.

**Corollary 2.** *The equation  $xyz + t^2 - 1 = 0$  has infinitely many integer solutions with all four coordinates nonzero.*

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*Proof.* For every integer  $n \notin \{0, 1, 2\}$ , Theorem 1 with  $(a, b, c) = (1, 1, n)$  and  $\varepsilon = 1$  gives

$$(x, y, z, t) = (1, 1, n(2 - n), 1 - n).$$

All four coordinates are nonzero, and the resulting points are distinct because their fourth coordinates are distinct.  $\square$

## 2. RATIONAL COMPLETENESS

The polynomial  $xyz + t^2 - 1$  is irreducible over  $\mathbb{Q}$ . Indeed, viewed as a polynomial in  $z$  over the unique factorization domain  $\mathbb{Q}[x, y, t]$ , it is primitive and has degree one; it is therefore irreducible by Gauss's lemma. Hence  $X_{\mathbb{Q}}$  is irreducible.

Let

$$U = \{(x, y, z, t) \in X(\mathbb{Q}) : xy \neq 0\}$$

and let

$$V = \{(a, b, c) \in \mathbb{A}^3(\mathbb{Q}) : ab \neq 0\}.$$

Both are dense open sets in their respective affine varieties; for example,  $(1, 1, 1, 0) \in U$ .

**Proposition 3.** *For either sign  $\varepsilon \in \{1, -1\}$ , the restriction*

$$\Phi_{\varepsilon}|_V : V \longrightarrow U$$

*is an isomorphism. Its inverse is*

$$\Psi_{\varepsilon}(x, y, z, t) = \left(x, y, \frac{1 - \varepsilon t}{xy}\right).$$

*Thus either  $\Phi_{\varepsilon}$  is a birational parametrization of  $X$ .*

*Proof.* If  $(a, b, c) \in V$ , then the first two coordinates of  $\Phi_{\varepsilon}(a, b, c)$  have product  $ab \neq 0$ , so the image lies in  $U$  by Theorem 1. Moreover, if  $t = \varepsilon(1 - abc)$ , then

$$1 - \varepsilon t = abc,$$

and therefore

$$\Psi_{\varepsilon}(\Phi_{\varepsilon}(a, b, c)) = \left(a, b, \frac{abc}{ab}\right) = (a, b, c).$$

Conversely, take  $(x, y, z, t) \in U$  and set

$$c = \frac{1 - \varepsilon t}{xy}.$$

Then  $xyz = 1 - \varepsilon t$ , and hence

$$\varepsilon(1 - xyz) = \varepsilon(\varepsilon t) = t.$$

For the third coordinate, the defining equation of  $X$  gives  $xyz = 1 - t^2$ , so

$$\begin{aligned} c(2 - xyz) &= \frac{1 - \varepsilon t}{xy}(2 - (1 - \varepsilon t)) \\ &= \frac{(1 - \varepsilon t)(1 + \varepsilon t)}{xy} \\ &= \frac{1 - t^2}{xy} = z. \end{aligned}$$

Thus  $\Phi_{\varepsilon}(\Psi_{\varepsilon}(x, y, z, t)) = (x, y, z, t)$ .  $\square$

**Corollary 4.** *The two polynomial maps  $\Phi_1$  and  $\Phi_{-1}$  together cover all of  $X(\mathbb{Q})$ .*

*Proof.* Every rational point with  $xy \neq 0$  is covered by either map by Proposition 3. Suppose instead that  $xy = 0$ . The defining equation then gives  $t^2 = 1$ , so  $t \in \{1, -1\}$ . Since 2 is invertible in  $\mathbb{Q}$ ,

$$(x, y, z, t) = \Phi_t \left( x, y, \frac{z}{2} \right).$$

Indeed,  $xy = 0$  implies that the final two coordinates on the right are  $2(z/2) = z$  and  $t(1 - 0) = t$ .  $\square$

### 3. THE EXACT INTEGRAL IMAGE

Rational completeness does not imply that every integer point is obtained from integer parameters. The precise obstruction is elementary divisibility.

**Proposition 5.** *Let  $P = (x, y, z, t) \in X(\mathbb{Z})$ , and fix  $\varepsilon \in \{1, -1\}$ .*

(1) *If  $xy \neq 0$ , then*

$$P \in \Phi_\varepsilon(\mathbb{Z}^3) \iff xy \mid 1 - \varepsilon t.$$

*In that case the unique parameter triple mapping to  $P$  is*

$$(a, b, c) = \left( x, y, \frac{1 - \varepsilon t}{xy} \right).$$

(2) *If  $xy = 0$ , then*

$$P \in \Phi_\varepsilon(\mathbb{Z}^3) \iff t = \varepsilon \text{ and } 2 \mid z.$$

*Proof.* Suppose first that  $xy \neq 0$ . Any preimage must satisfy  $a = x$  and  $b = y$ . The fourth coordinate then forces

$$abc = 1 - \varepsilon t, \quad \text{hence} \quad c = \frac{1 - \varepsilon t}{xy}.$$

Thus integrality of  $c$  is necessary. It is also sufficient: if the stated divisibility holds, choose this integer  $c$ . The computation in the proof of Proposition 3, using  $xyz = 1 - t^2$ , shows that the resulting third coordinate is exactly  $z$ . Uniqueness follows from the displayed formula for  $c$ .

Now suppose that  $xy = 0$ . Again  $a = x$  and  $b = y$ , so  $abc = 0$ . Therefore

$$\Phi_\varepsilon(a, b, c) = (x, y, 2c, \varepsilon).$$

This equals  $P$  precisely when  $t = \varepsilon$  and  $z = 2c$  for some  $c \in \mathbb{Z}$ , which is equivalent to  $t = \varepsilon$  and  $2 \mid z$ .  $\square$

*Remark 6.* Neither chart, nor their union, is surjective on  $X(\mathbb{Z})$ . For example,

$$(3, 4, -2, 5) \in X(\mathbb{Z}),$$

because  $3 \cdot 4 \cdot (-2) + 5^2 - 1 = -24 + 24 = 0$ . However,  $xy = 12$  divides neither  $1 - t = -4$  nor  $1 + t = 6$ . By Proposition 5, this point lies in neither  $\Phi_1(\mathbb{Z}^3)$  nor  $\Phi_{-1}(\mathbb{Z}^3)$ .

Since the equation is symmetric in  $x, y, z$ , one may instead place  $c(2 - abc)$  in either of the other two coordinates. This gives the two corresponding permuted chart types for each sign.